ICDFA GRC102

WEEK 5 LINUX SECURITY GOVERNANCE LAB

Professional Lab Report & Evidence

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Course / Module	GRC102
Submission	Week 5 Linux Security Governance Lab
Date	7TH OCTOBER 2026

1. Introduction

This report documents the practical activities completed for the Week 5 Linux Security Governance Lab. The work focuses on establishing a simulated environment, identifying security governance weaknesses, implementing governance controls, and developing metrics and reporting mechanisms.

Screenshots are intentionally represented as labelled placeholders. The completed screenshots can be inserted in the indicated locations without reproducing terminal commands or scripts in the report.

2. Lab Environment Overview

The practical work used a simulated Linux-based environment to demonstrate security governance activities, vulnerability management, control implementation, verification, monitoring, and executive reporting. The evidence sections below are arranged in the required Part 1 → Part 4 order.

Part 1 — Setting Up a Simulated Environment

1.1 Docker-Based Environment

The simulated environment was prepared using Docker to provide the required systems and services for the lab activities. The environment was checked to confirm that the required components were available and running.



[Screenshot – Part 1.1: Docker-based environment / running containers]

1.2 Vulnerability Scanning

A vulnerability scanning activity was performed against the simulated environment to identify weaknesses that could affect the security posture of the systems.



[Screenshot – Part 1.2: Vulnerability scanning activity / result]

1.3 Patch Management

Patch management activities were carried out to identify and address system vulnerabilities. The results provide evidence of the organisation's ability to track and improve patch compliance.



[Screenshot – Part 1.3: Patch management activity / result]

1.4 Governance Tracking

Governance tracking was established to provide visibility into security issues, ownership, status, and remediation activities.



[Screenshot – Part 1.4: Governance tracking]

1.5 Initial Governance Data

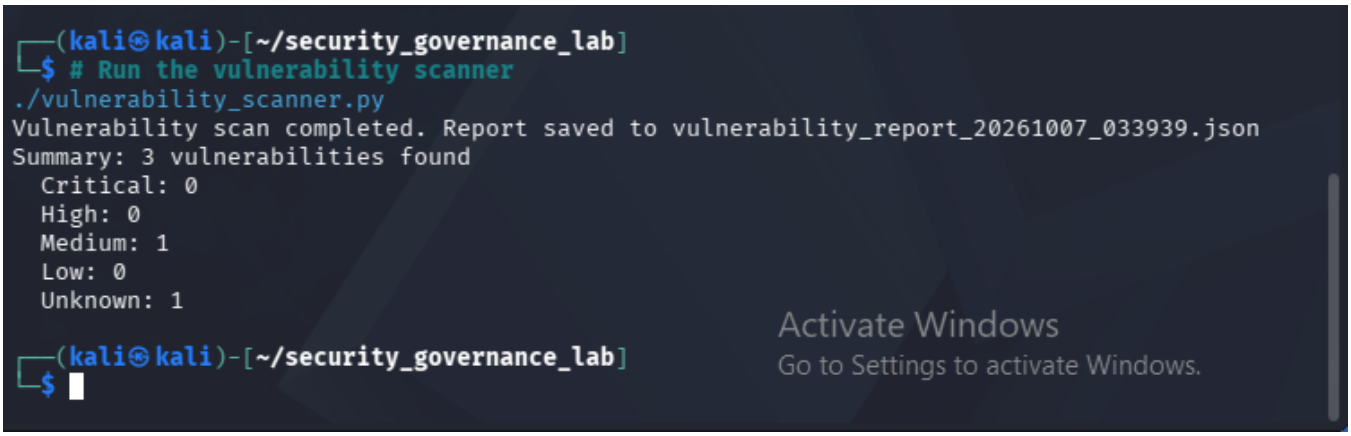
Initial governance data was recorded to establish a baseline for monitoring security risks, vulnerabilities, controls, and remediation status.



[Screenshot – Part 1.5: Initial governance data]

1.6 Vulnerability Scan Report

The vulnerability scan results were reviewed and documented to support risk identification and subsequent governance decisions.



[Screenshot – Part 1.6: Vulnerability scan report]

Part 2 — Analyzing Security Governance Failures

2.1 Attack Simulation

An attack simulation was performed against the simulated environment to demonstrate how weaknesses can be exploited when security controls and governance processes are ineffective.



[Screenshot – Part 2.1: Attack simulation]

2.2 Governance Failure Analysis

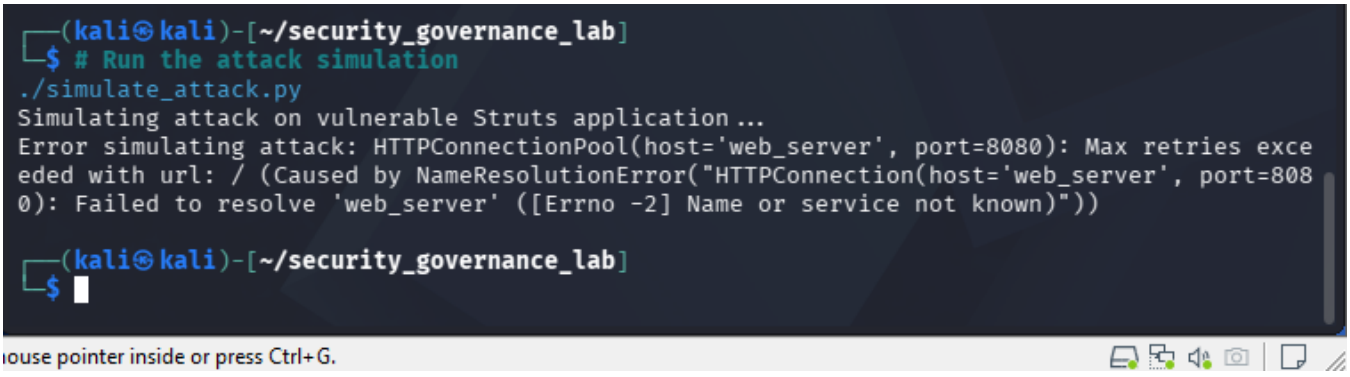
The observed results were analysed from a governance perspective, focusing on weaknesses in accountability, vulnerability management, control implementation, monitoring, and management oversight.



[Screenshot – Part 2.2: Governance failure analysis]

2.3 Equifax Comparison

The simulated governance failures were compared with lessons from the Equifax case study. The comparison highlights how weaknesses in vulnerability management, asset visibility, control assurance, and oversight can increase organisational risk.



[Screenshot – Part 2.3: Equifax comparison / analysis]

Key Governance Failures and Lessons Learned

Governance Failure	Lesson Learned
Weak vulnerability and patch management	Critical vulnerabilities require timely identification, ownership, remediation, and verification.
Insufficient network segmentation	Segmentation can reduce the impact and spread of a compromise.

Weak database protection	Sensitive data requires appropriate access controls, monitoring, and security safeguards.
Insufficient security monitoring	Continuous monitoring improves detection, response, and governance visibility.
Limited executive oversight	Boards and executives need meaningful security metrics and clear accountability.

Part 3 — Implementing Security Governance Controls

The controls below were implemented to address the governance weaknesses identified during the analysis. Evidence placeholders are provided for the corresponding implementation and verification screenshots.

3.1 Patch Management Control

A patch management control was implemented to improve the organisation's ability to identify, prioritise, apply, and verify security updates.



[Screenshot – Part 3.1: Patch management control]

3.2 Network Segmentation Control

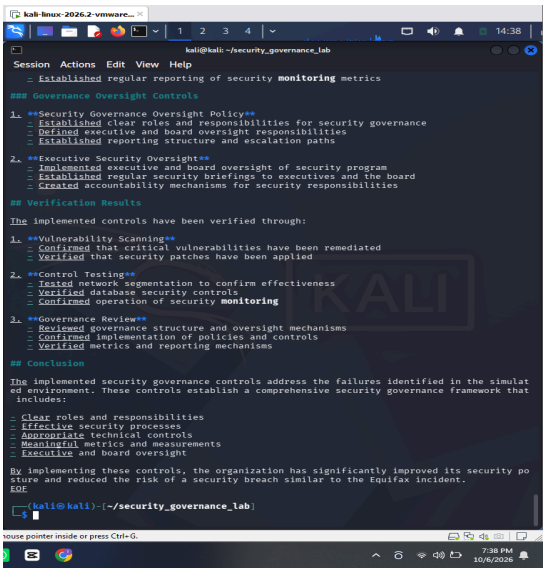
Network segmentation was implemented to limit unnecessary communication between systems and reduce the potential impact of a security compromise.



[Screenshot – Part 3.2: Network segmentation control]

3.3 Database Security Control

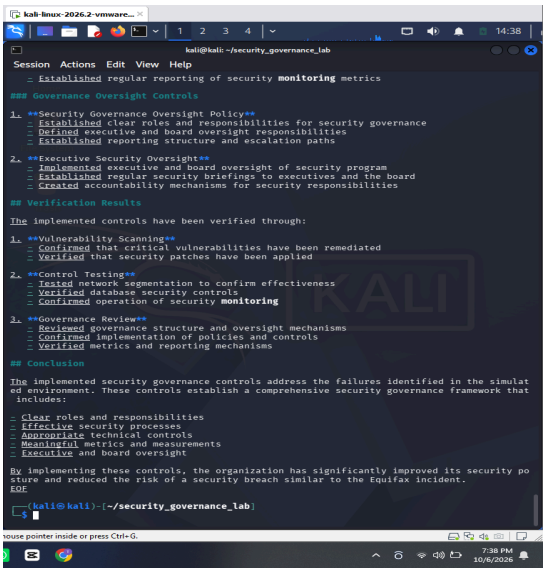
Database security controls were implemented to strengthen protection of sensitive information and restrict unauthorised access.



[Screenshot – Part 3.3: Database security control]

3.4 Security Monitoring Control

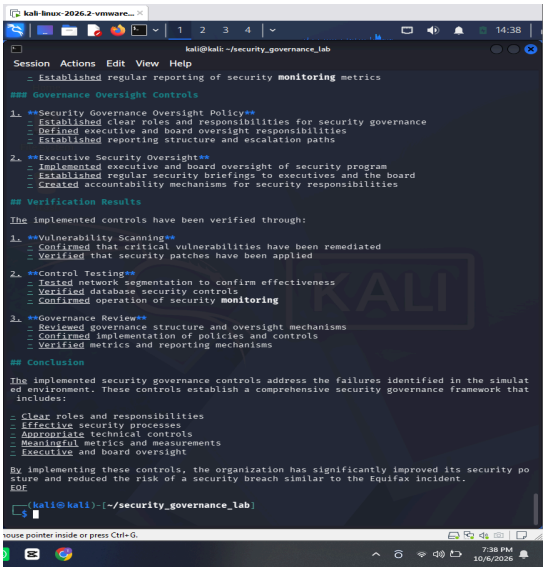
Real-time security monitoring was implemented to improve visibility into security events and support timely detection and response.



[Screenshot – Part 3.4: Security monitoring control]

3.5 Governance Oversight Control

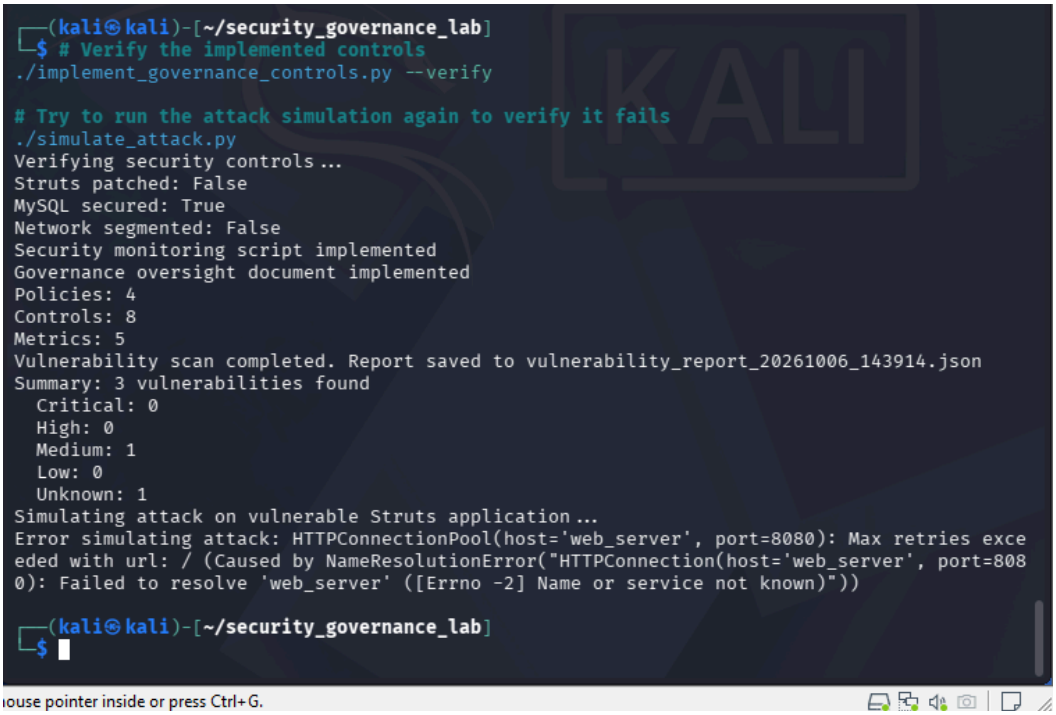
Governance oversight mechanisms were established to provide management and executive stakeholders with visibility into security posture, risks, control performance, and remediation.



[Screenshot – Part 3.5: Governance oversight control]

3.6 Verification of Controls

The implemented controls were verified to confirm that they were functioning as intended. Verification also included rerunning the relevant security/attack simulation activities where required.



[Screenshot – Part 3.6: Control verification / rerun results]

How the Implemented Controls Address the Governance Failures

The implemented controls directly address the weaknesses identified in the simulated environment and the Equifax comparison. Patch management improves vulnerability remediation, segmentation reduces the potential spread of compromise, and database security strengthens protection of sensitive information. Continuous monitoring provides better visibility into security events, while governance oversight improves accountability and ensures that security risks and control performance receive appropriate management attention.

Together, these controls move security governance from reactive identification of weaknesses toward a more structured approach based on prevention, monitoring, verification, accountability, and continuous improvement.

Part 4 — Developing Security Governance Metrics and Reporting

4.1 Security Governance Metrics

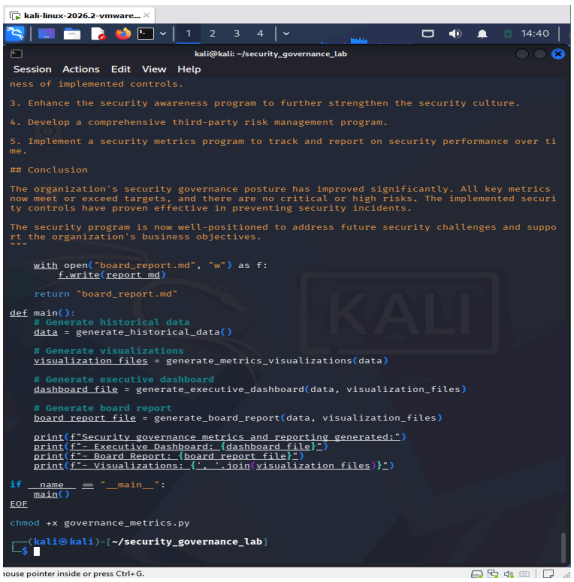
Security governance metrics were defined to provide measurable visibility into the effectiveness of key controls and the organisation's security posture.



[Screenshot – Part 4.1: Security governance metrics]

4.2 Executive Dashboard

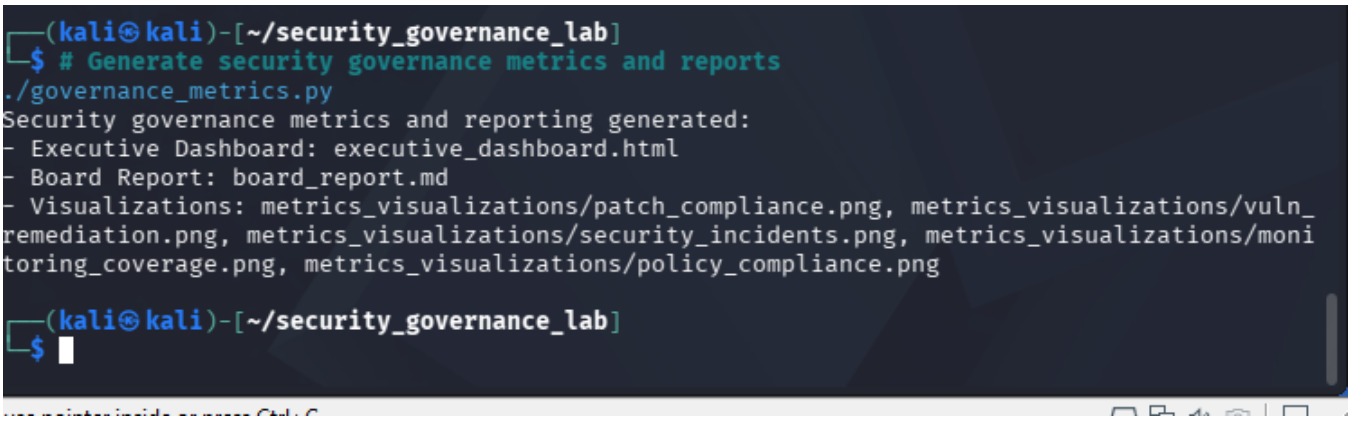
An executive dashboard was developed to present key security governance indicators in a concise format suitable for management and executive review.



[Screenshot – Part 4.2: Executive dashboard]

4.3 Metrics Visualizations

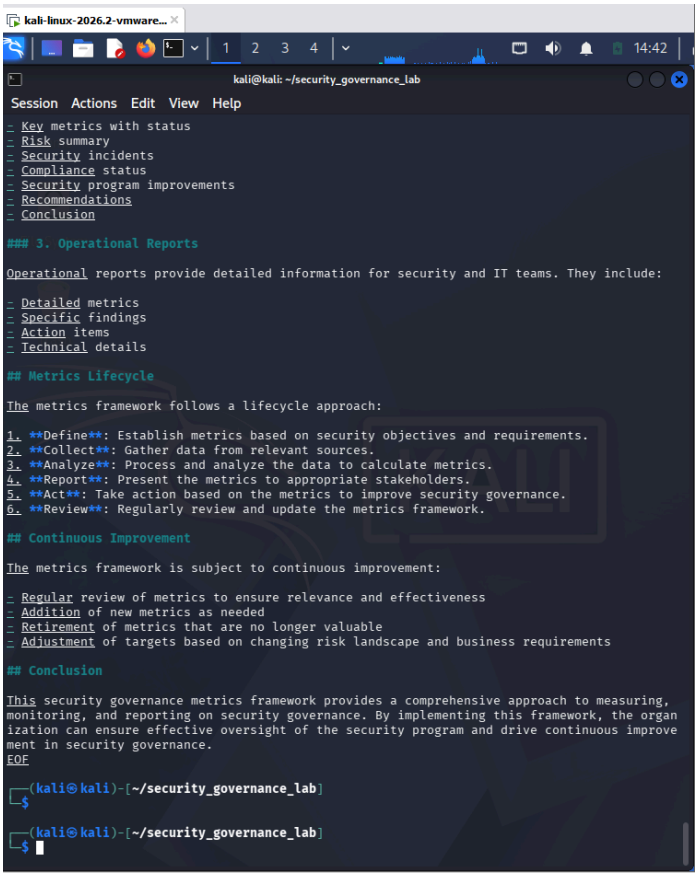
Visualisations were used to make trends, performance levels, risks, and areas requiring attention easier to understand.



[Screenshot – Part 4.3: Metrics visualizations]

4.4 Board Report

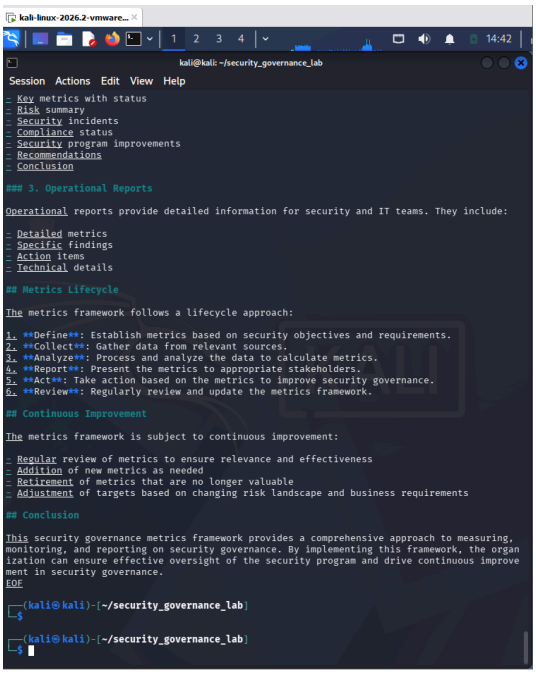
A board-level report was prepared to communicate security posture, significant risks, control performance, and areas requiring management attention.



[Screenshot – Part 4.4: Board report]

4.5 Metrics Framework

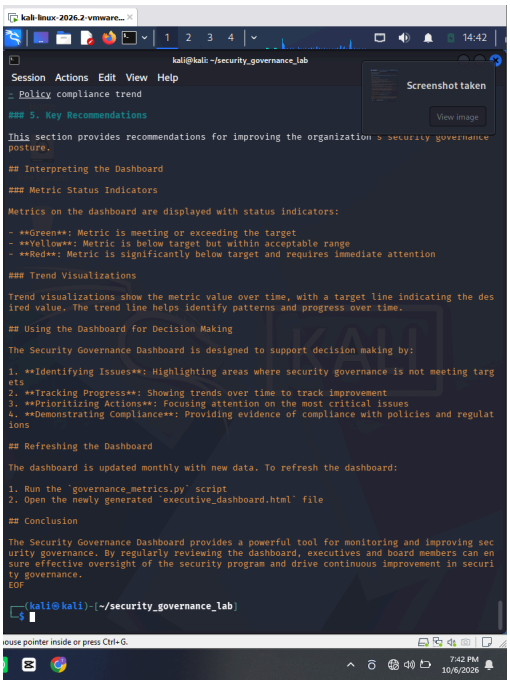
A metrics framework was documented to define the purpose, measurement approach, ownership, reporting frequency, and interpretation of the selected governance metrics.



[Screenshot – Part 4.5: Metrics framework]

4.6 Dashboard User Guide

A user guide was prepared to explain how the dashboard should be accessed, interpreted, and used for governance decision-making.



[Screenshot – Part 4.6: Dashboard user guide]

Security Governance Metrics Framework

Metric	Purpose	Governance Value	Reporting Focus
Patch Compliance	Measure the proportion of systems meeting patch requirements.	Shows effectiveness of vulnerability management.	Compliance trend and exceptions
Vulnerability Remediation Time	Track how quickly identified vulnerabilities are addressed.	Shows remediation effectiveness and risk exposure.	Average/overdue remediation
Security Incidents	Track security incidents and their trends.	Supports risk monitoring and response oversight.	Incident volume and severity
Monitoring Coverage	Measure the extent of systems and assets covered by monitoring.	Shows visibility into the environment.	Coverage gaps and improvement
Policy Compliance	Measure adherence to approved security policies and requirements.	Supports accountability and control assurance.	Compliance rate and exceptions

Metrics Lifecycle

The metrics lifecycle consists of defining relevant measures, collecting reliable data, validating the data, analysing trends, reporting results to the appropriate stakeholders, taking corrective action, and reviewing the metrics periodically to ensure they remain useful for governance.

How Metrics and Reporting Support Effective Security Governance

Security metrics and reporting provide management with measurable evidence of how well security controls and governance processes are performing. Rather than relying only on technical details, the selected metrics help translate security activity into information that can support risk-based decisions, prioritisation, accountability, and resource allocation.

Regular reporting also supports continuous improvement by making trends, exceptions, unresolved vulnerabilities, incidents, monitoring gaps, and policy compliance visible to the appropriate stakeholders. This enables management and the board to challenge performance, monitor remediation, and maintain oversight of the organisation’s security posture.

Screenshot Evidence Index

☐ Part 1.1 — Docker-based environment

☐ Part 1.2 — Vulnerability scanning

☐ Part 1.3 — Patch management

☐ Part 1.4 — Governance tracking

☐ Part 1.5 — Initial governance data

☐ Part 1.6 — Vulnerability scan report

☐ Part 2.1 — Attack simulation

☐ Part 2.2 — Governance failure analysis

☐ Part 2.3 — Equifax comparison

☐ Part 3.1 — Patch management control

☐ Part 3.2 — Network segmentation control

☐ Part 3.3 — Database security control

☐ Part 3.4 — Security monitoring control

☐ Part 3.5 — Governance oversight control

☐ Part 3.6 — Control verification

☐ Part 4.1 — Security governance metrics

☐ Part 4.2 — Executive dashboard

☐ Part 4.3 — Metrics visualizations

☐ Part 4.4 — Board report

☐ Part 4.5 — Metrics framework

☐ Part 4.6 — Dashboard user guide

Conclusion

This report provides a structured record of the Week 5 Linux Security Governance Lab activities. The practical work demonstrates the relationship between technical security controls, governance responsibilities, continuous monitoring, assurance, and executive reporting.

Before submission, insert the completed screenshots into their corresponding labelled locations and confirm that all evidence is readable and arranged in Part 1 → Part 4 order.